

Members of the team

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Roles of this document

1. George Douroukas (project manager, writer)
2. Tasos Venos (check)

Tools

1. Google Doc
2. Visual Paradigm
3. Microsoft Word

(a)

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(b)

Below is an initial scheduling and estimation of work represented by the respective Gantt chart and Pert chart. Gantt chart: this is a chart that helps teams to plan their work based on deadlines and allocate their resources properly. The figure depicts the relationship between start and end date of tasks and pieces that depend on other pieces. Decisions on how to sequence the deliverables, the pieces of each deliverable and their order were made through group discussions with the aim of achieving consensus among all members and the course schedule. It was also discussed which pieces might need to be rebuilt due to a forecast error which will be seen in practice. There will follow 4 figures, 3 showing the deliverables in detail with focus and better resolution and 1 figure showing how they have been represented overall. The aim is to help make them easy to read and understand.

In figure 1 we can see the first 2 deliverables with their requests, their start and end date, and see how long the task and each task lasts. At the end of the task there is a diamond indicating its completion date. Finally, the fact that the bars in the first deliverable look darker than the second is because it is 100% implemented as opposed to the second one, where implementation has not started.

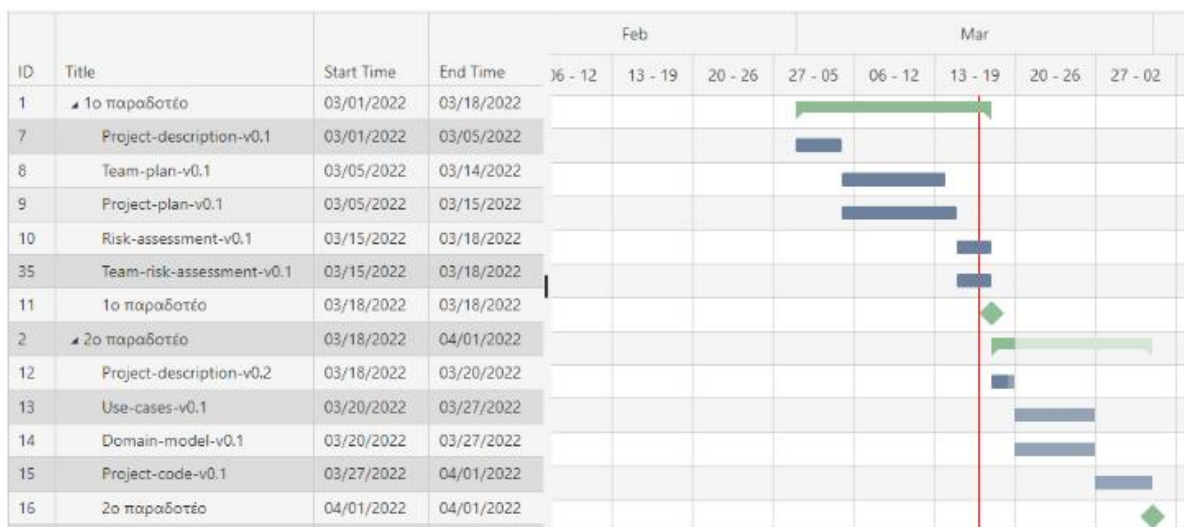


Figure 1: Gantt Chart

Similarly, we see the 3rd, 4th deliverable, with their dates, but in pale colours as they have not been implemented.

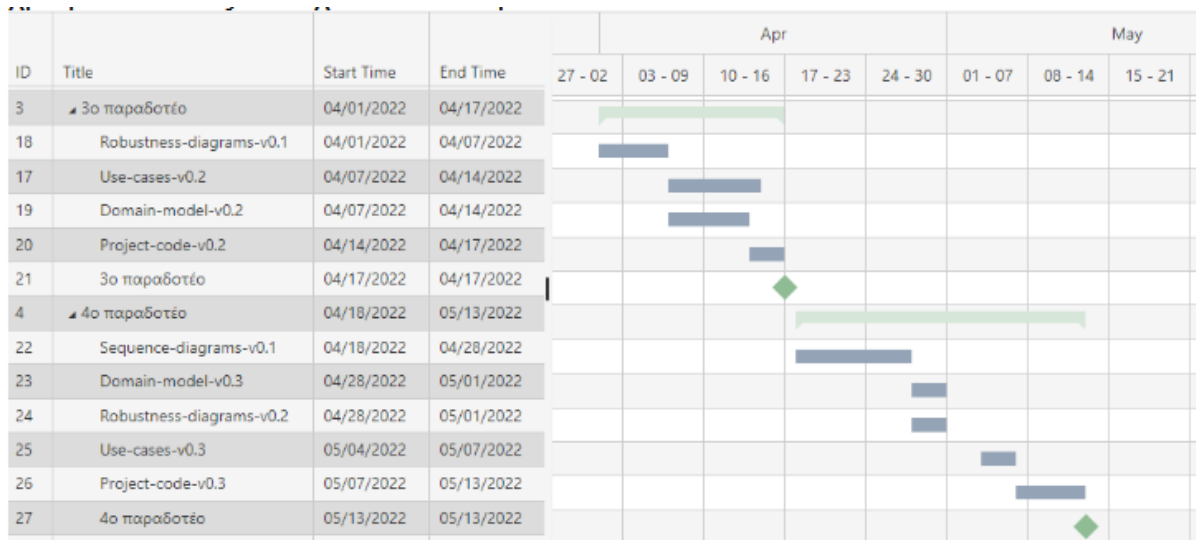


Figure 2: Gantt Chart

The following is the 5th,6th deliverable with the same indications as before

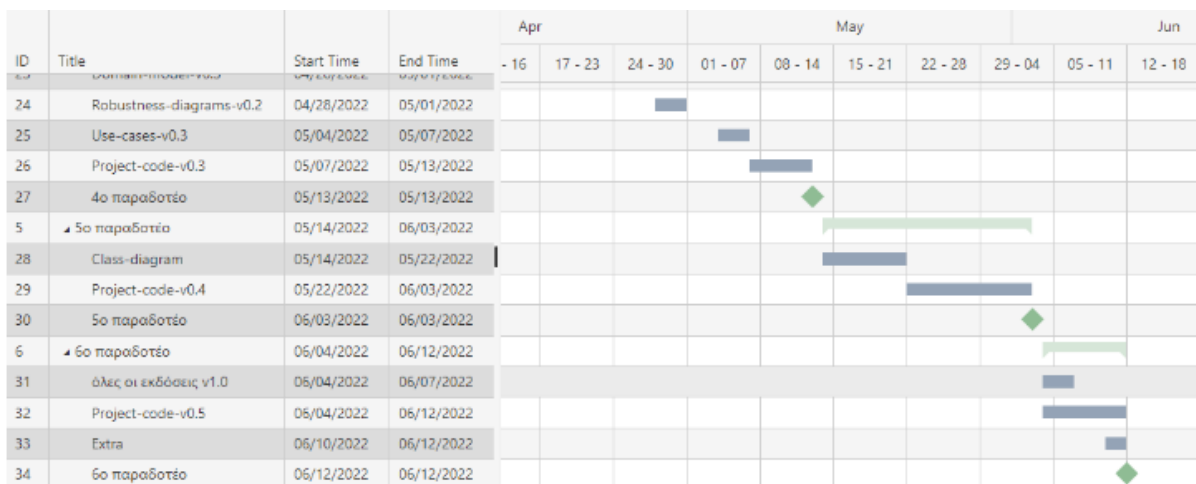


Figure 3: Gantt Chart

Finally, the whole Gantt chart of the project is mapped. The image came out quite large so it was not clearly visible, so the photos were given fragmented focus and the overall one was placed at the end to show a picture of what it looks like.

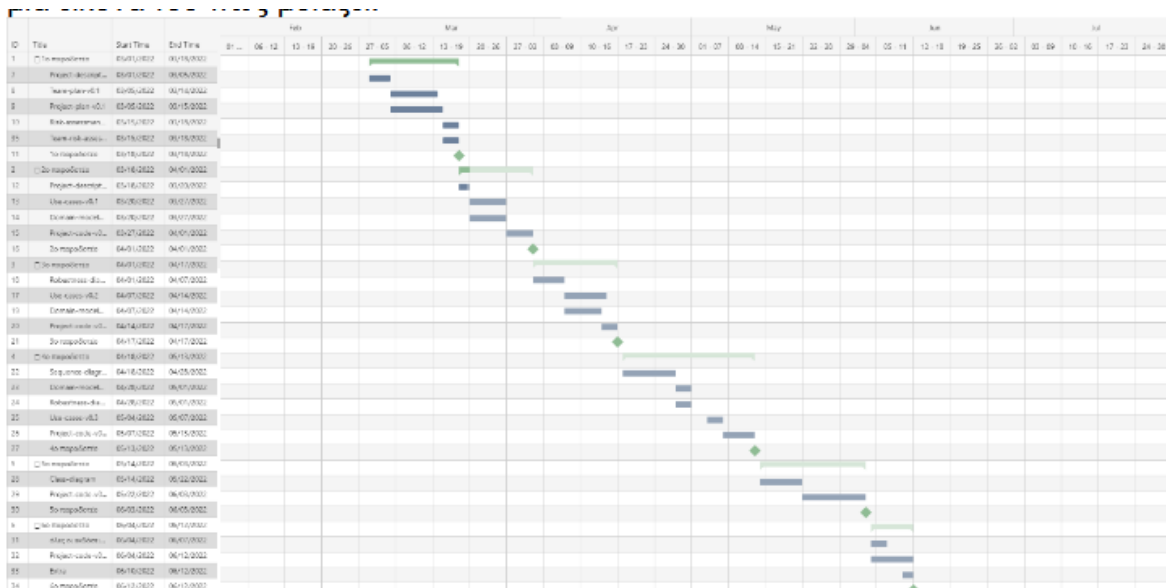
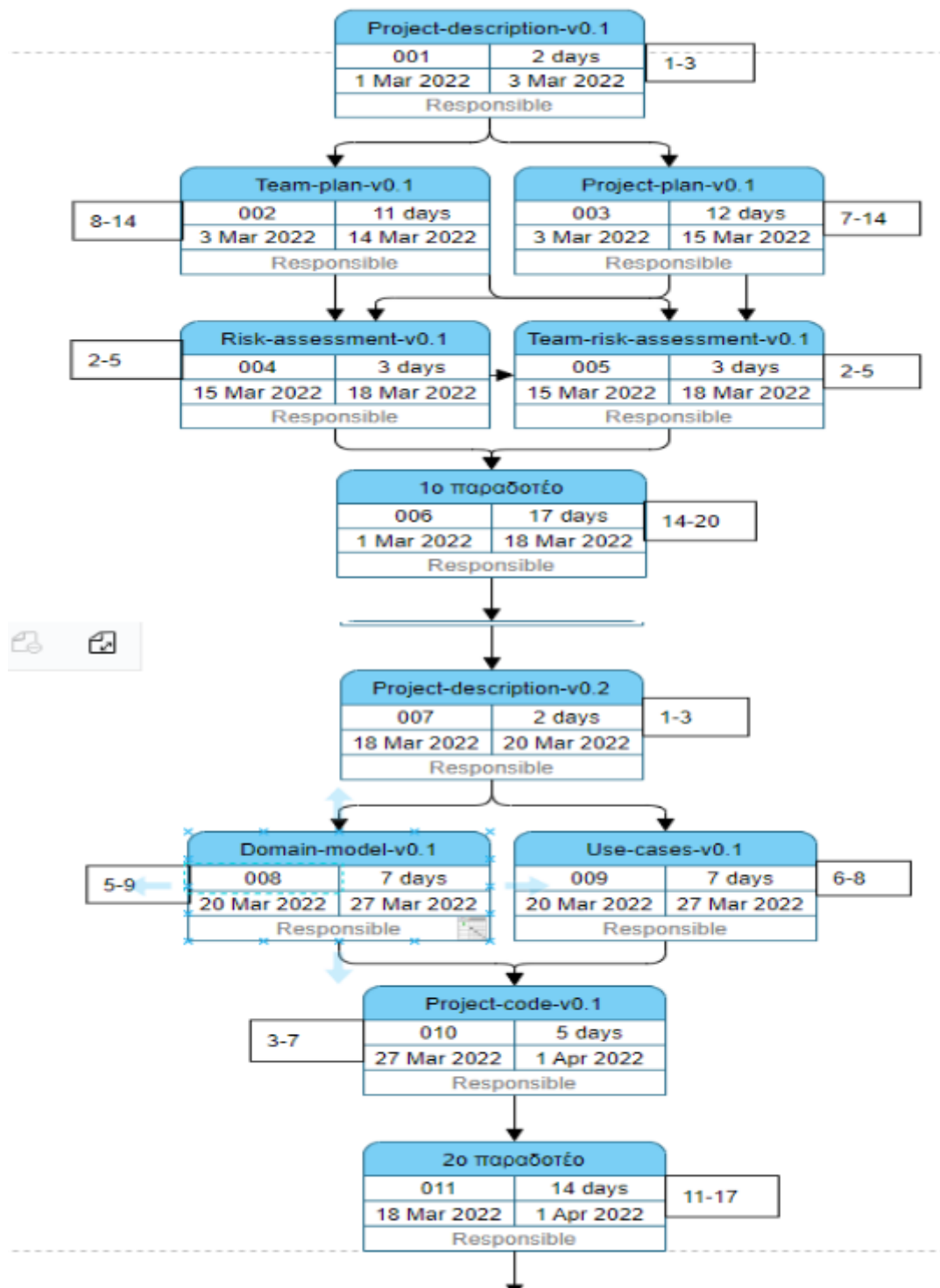


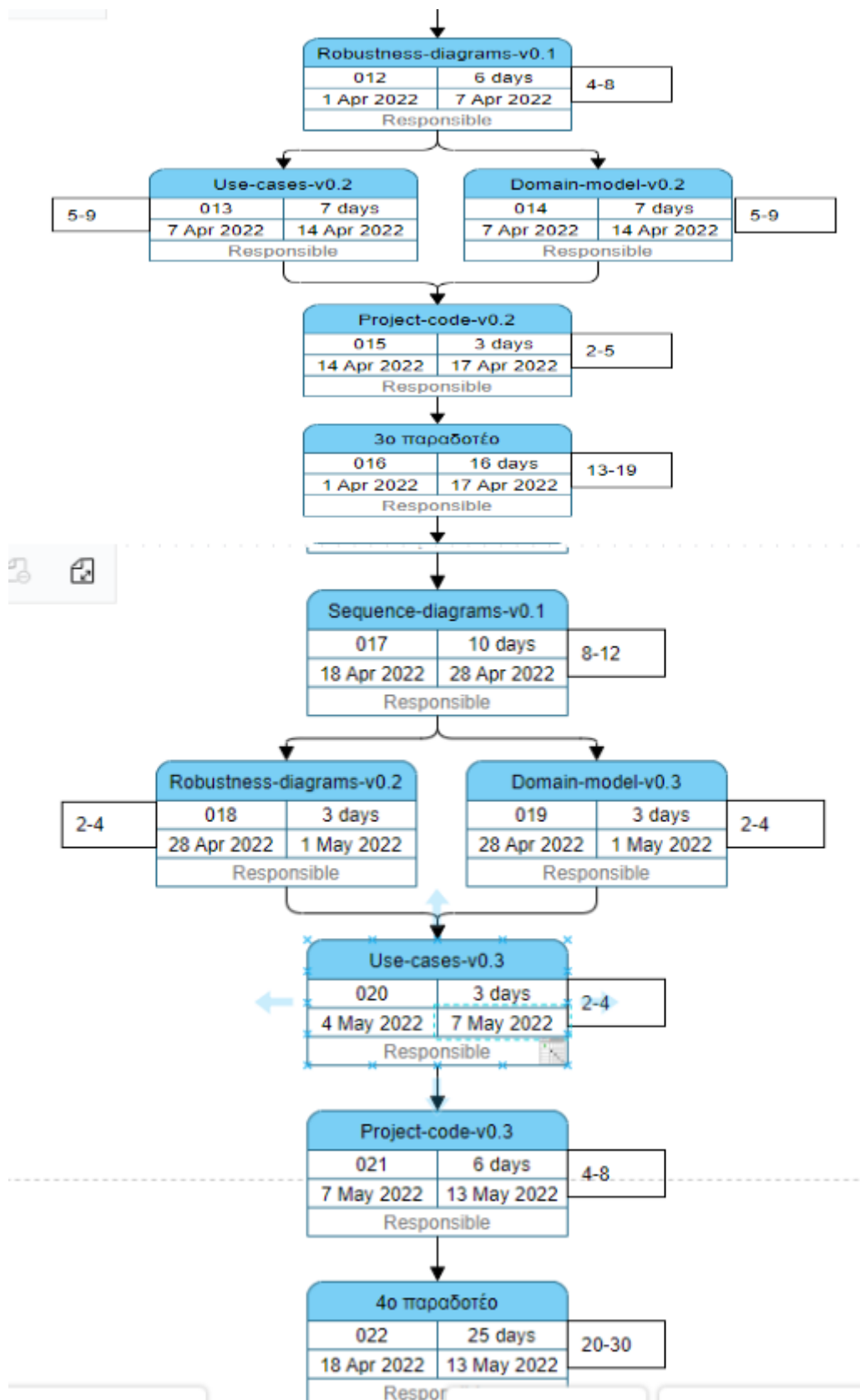
Figure 4: Gantt Chart

Pert chart: This chart identifies the critical path for estimating the minimum time it will take to complete all the tasks in a project. This is done by examining the project breakdown, estimating the duration of each activity, the dependencies between them and the order in which they need to be completed. Two further dates are also set, which are the optimistic date and the pessimistic date. The optimistic date is calculated on the basis that if no problems arise between the execution of the requested tasks, and the cooperation of the members is excellent then it is expected that the project will be finished earlier than normal. The pessimistic one is due to cooperation and/or communication problems and may cause the project to be delayed. The Pert chart of the project as a whole in pieces will follow below to better focus on the date and sequence dependencies. We start with the first piece of each deliverable which if it is alone means that if it is not completed the others do not start, if it is 2 or more in parallel means that they are executed simultaneously. When completed, the projects of each deliverable end up in a cell with its name to indicate that it has been completed.

Each end of a deliverable marks the start of the new pieces of the next deliverable to complete it. The tool that created Pert did not have a diamond to show it, so an assumption is made and instead of a diamond, completion is shown with a cell. In the blue box the name of each requested deliverable is written. Just below that are 4 stacked cells and a cell to the left or right of the box that is unstacked. In the top left cell is written the number of the node that simply numbers the requested item (1,2,3 etc.). In the top right cell is written the estimated days it will take to implement the project. In the bottom left cell, the start date and in the bottom right cell the end date. The unordered box shows the optimistic and pessimistic durations discussed above. It contains a type 1-3 number meaning that the optimistic date is 1 and the pessimistic date is 3. Although several images follow, I am treating it as 1 with no

separate designations so that it has continuity and looks as if it normally did in the implemented program.





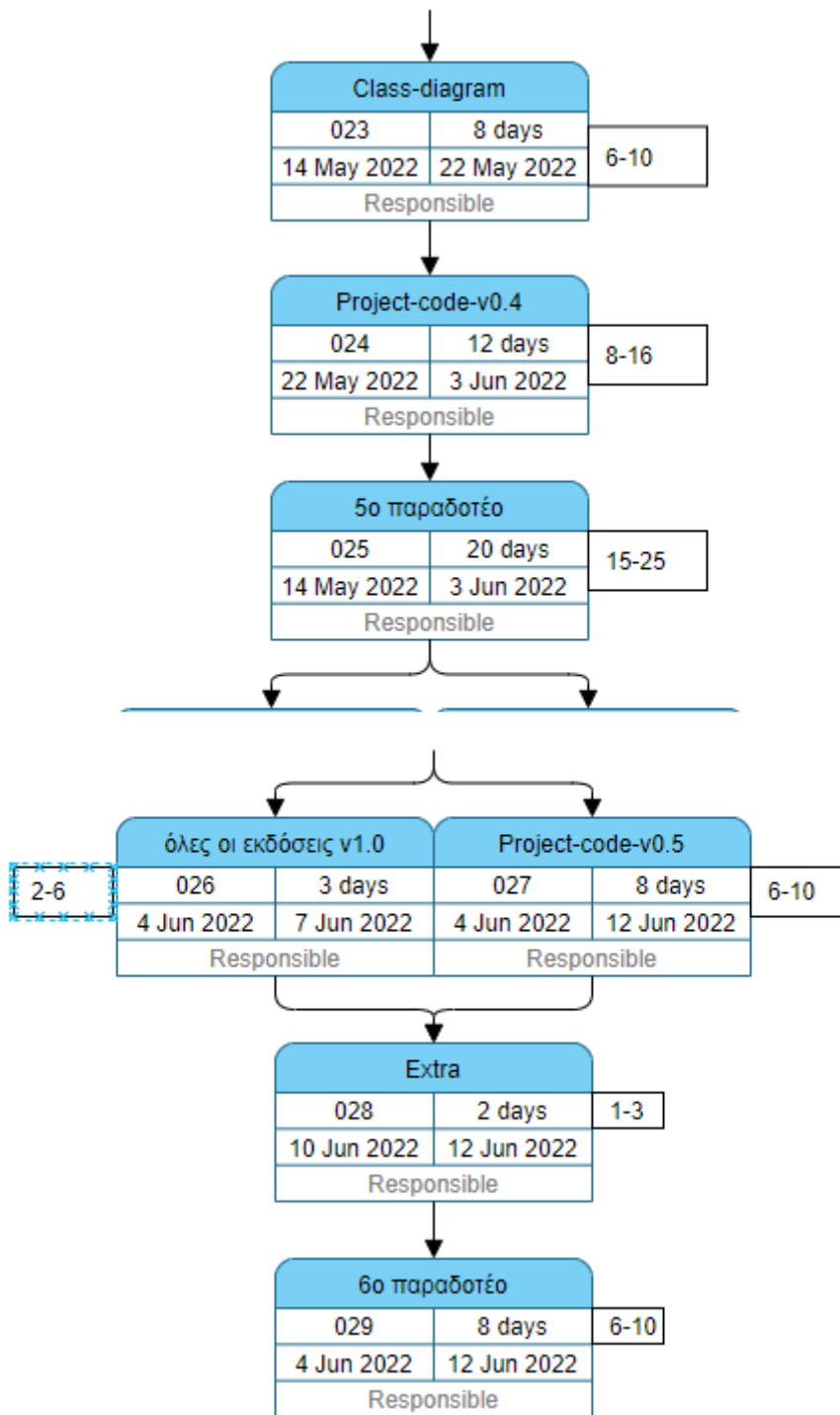


Figure 5: PERT Chart

(c)

The group during its creation and afterwards held online meetings using a communication platform (discord) as there is no possibility of physical presence due to the distance between the members. In the initial phase each member proposed ideas for the topic of the project and what was acceptable to all members was decided to be carried out. Then each member took up a piece of deliverable, for workload sharing and quick result. So, the Kanban function was adopted, but meetings were held, which is a Scrum function, to discuss and check each member with the others to see what has been accomplished. Another reason for a meeting is that some deliverables are interdependent and need communication from the whole team for their realization. Before each delivery of a piece of the project, each member gives the piece they implemented to a member and vice versa to check for errors. Since the team consists of 4 people, two pairs are created to exchange tasks for checking, each time formed differently (having a' b' c' d' students, the first one will check the first one, then a' will check the second one and vice versa, then a' will check the third one, etc.) Before the delivery, the team checks the deliverables together, to correct the smallest details.

(d)

Google doc: used to collect all deliverables in one file so that all team members can access it for updating, modifying and corrections. Visual-Paradigm: this is an application in which you can draw Gantt charts and Pert charts with ease providing many ways to design.

Microsoft word: create a deliverable and convert it to a Pdf file for delivery.

Microsoft Paint: create mock-up screens.

e) Programming language of the project: Python or Java